

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHONOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8503**

AV:
2025-26
Semi:
1
Exam Type:
Written

Session:
Odd MSE October 2025
Slot:
10:00am - 11:00am
Block: 401
Course Section: 1

Date:
08-10-2025
C. Code:
CS413
Seat No:
2022600039

L_29_496_986_1_1505_38463

Name of the candidate. Deep Patel

Candidate's UID number:

2	0	2	2	6	0	0	0	3	9
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Examination class room number:

4	0	1
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Invigilator's signature with date:

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 8/10/25

(To be filled in by the candidate)

EXAMINATION: B.E BTECH MSE
(For example FY MCA/FY BTECH)

PROGRAM: Deep CSE: AIML

DAY AND DATE: 8/10/25 Wed

COURSE NAME: Deep Learning

SEMESTER : 7

I / II / III / IV / V / VI / VII / VIII

(To be filled in by the examiner)								
Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Q22) In a depthwise convolution of size $H \times W \times C$, we

$$\text{loss fn} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Q23a) For a given RNN:

$$a_t = \sigma(W_{th}h_{t-1} + W_{tx}x_t + b_a)$$

$$a_t = \sigma(W_{th}h_{t-1} + W_{tx}x_t + b_a)$$

$$h_t = \tanh(a_t)$$

$$o_t = W_{oy}h_t + b_o$$

$$y_t = \tanh(o_t)$$

$$\therefore \frac{\partial L}{\partial h} = \frac{\partial \tanh(W_{oy}h_t + b_o)}{\partial h}$$

$$= \tanh(\sigma(W_{th}h_{t-1} + W_{tx}x_t + b_a))$$

The sigmoid function restricts the value between (0,1) during back propagation leading to a vanishing gradient.



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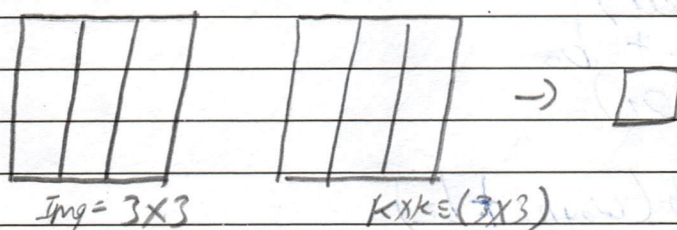
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Q2) a) For a standard ~~2D~~ 2D convolution, with $(H \times W)$ dimensions, we will have the dimension of the final output to be the following when we use a filter k for it:

$$H_o = \frac{H - F + 2P + 1}{S} \quad W_o = \frac{W - F + 2P + 1}{S} \quad \begin{matrix} P \rightarrow \text{padding} \\ S \rightarrow \text{stride} \end{matrix}$$

- we know for each unit of ~~width~~ H or W , we will be multiplying $(k \times k)$ times.

↳



- As we can see to get ~~one image~~ a final output of 1×1 , we needed $k \times k$ multiplications for a single cell.

Therefore,

Total number of multiplications can be given as:

$$n(\text{multiplications}) = (H_o' \times W_o')$$

$$\text{where } H_o' = \frac{H - F + 2P + 1}{S} \times (F \times F)$$

$$W_o' = \frac{W - F + 2P + 1}{S} \times (F \times F)$$

$$n(\text{multiplication}) = H_o \times W_o \times F^2 \text{ for a single filter.}$$

$$\text{For } k \text{ filters, } n(\text{multiplication}) = H_o \times W_o \times F^2 \times k.$$



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In a 3D convolutional network, we have dimensions as follows:

$$\text{Input size} = (H, W, C) \quad \text{Kernel size} = (F, F, C)$$

For each kernel, we get an output in the form of 2D Matrices.

Formula for H_o & W_o remaining the same.

~~H_o~~

However, the number of multiplication required for a unit is given as: $(F \times F \times C)$

∴ For K kernels & a 3D sized initial input,

$$n(\text{multiplication}) = H_o \times W_o \times (F^2 C) \cdot K.$$

⇒ 3D convolutions network prove to be of great use because of their following nature characteristics:

- Weight Sharing.
- Sparse Connectivity.

- They are generally less dense when compared to a similar ANN network; hence are better at computational efficiency and Training.

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Q3)b) We know in GRU, we have combined the input and forget as the ~~update gate~~ reset gate.

- This leads to there being fewer number of parameters in GRU compared to LSTM.

$$\begin{aligned} \text{No. of Parameters (GRU)} &= 3 \times (d_x \cdot d_h + d_h \cdot d_h + d_h) \\ \text{No. of Parameters (LSTM)} &= 4 \times (d_x \cdot d_h + d_h \cdot d_h + d_h) \end{aligned}$$

- The decrease in the number of trainable parameters results in a better chance of preventing overfitting.
- Overfitting occurs when model performs exceptionally well on training data, but not so in the testing data.
- This occurs when model is too complex and tends to learn from the noise in ~~test~~ training data.
- When the number of parameters decrease, we tend to have a model which ~~works~~ has lesser overfitting.
- Thus GRUs tend to have better generalization.
- GRUs prove to be of great use in NLP application like language translations. Combined with the fact that they have better generalization & are less resource intensive than LSTM; they prove to be used in locally runnable models of Google Translate.



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Q2b) Conditions for auto encoder to be equivalent to PCA:

- The encoder and decoder functions must be linear.
- We must use a undercomplete encoder.

⇒ Overcomplete encoder is one where in the dimensions of the hidden state are more than the input state.

- This means while training, the hidden layer may simply copy the input state's initialization.
- This would lead to overfitting.

- To prevent this, we use regularization techniques like L2 regularization.
- It penalizes the large weights thus preventing overfitting.

Q1a) $L = \frac{1}{2} \|Xw - y\|^2$

∴ $\frac{\partial L}{\partial w} = \|Xw - y\| \cdot X$

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Rough work. *$\frac{1}{2} at =$* 